

Logistic Regression

Prof. Julie Lee

November 5, 2024

Link for downloading slides: www.sunju-lee.com/demo

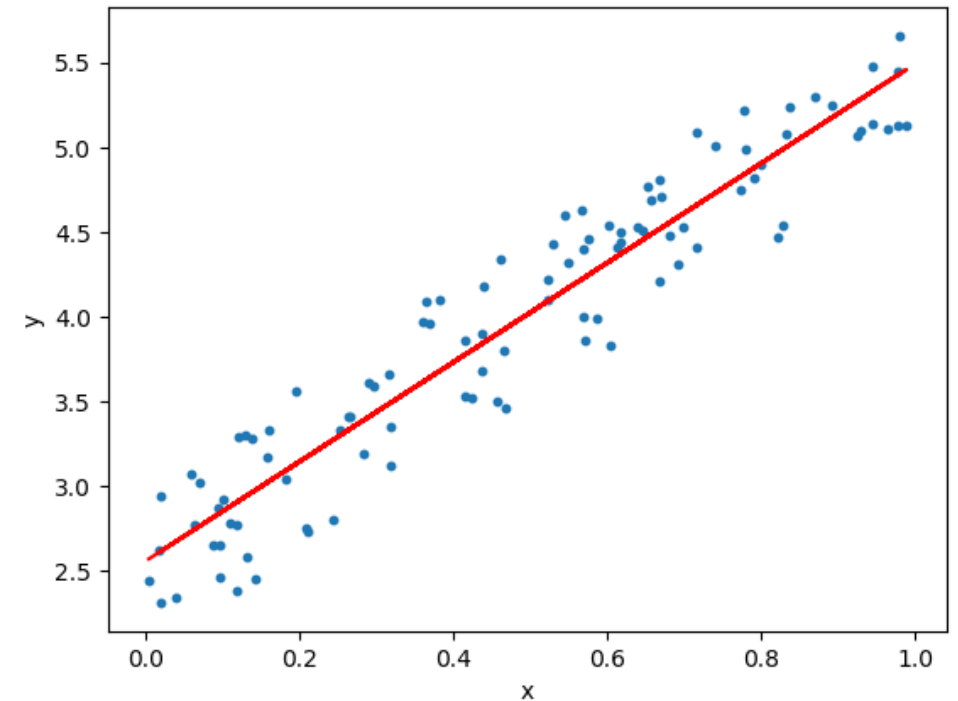
Today, we'll learn...

- Examples of logistic regression in the world
- What is logistic regression and what does the model look like?
- How can we perform classification using logistic regression?

Review of (simple) linear regression

In linear regression, we want to predict a *continuous* outcome by fitting a linear model to the data.

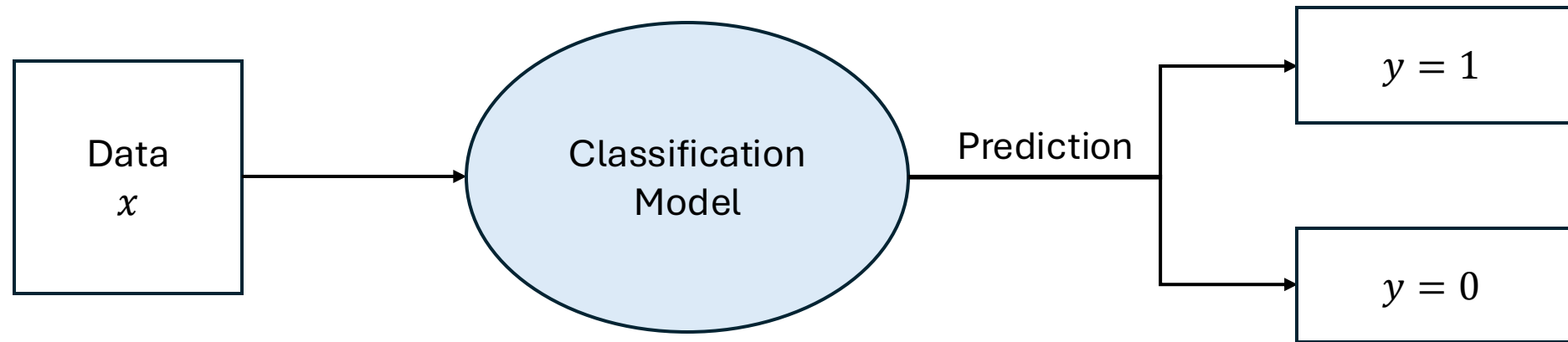
Example:



What if we want to predict a categorical outcome?

- Example:
- **Classification** is the task of predicting which class new data belongs to.
- In classification, we try to predict an outcome which has *classes*:
 - Ex. Binary classification: Yes ($y=1$) / No ($y=0$)
 - Ex. Multi-class classification: Dog/Cat/Horse

(Binary) Classification



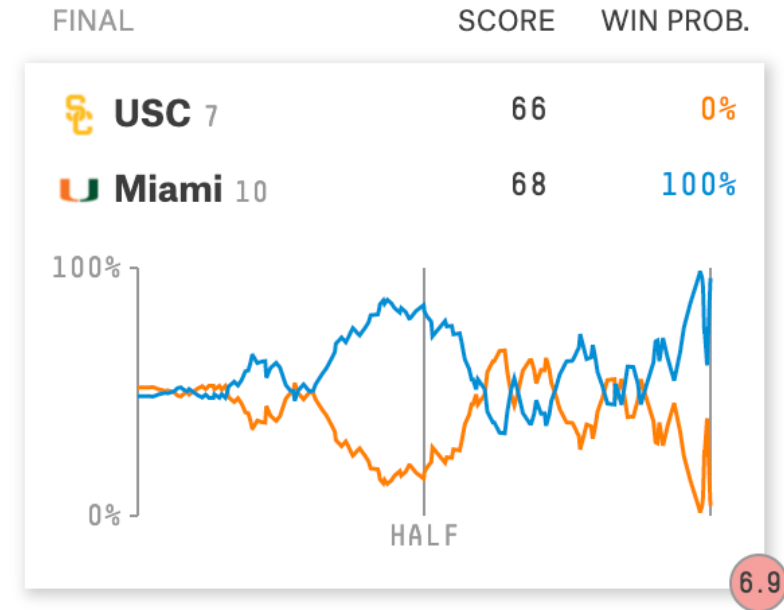
Logistic regression in the world:

1. Diabetes prediction
2. Adverse perinatal outcomes prediction (stillbirth, preterm birth)
3. Concussion diagnosis



Logistic regression in the world:

- 4. Fraud detection in financial transactions
- 5. Spam detection in emails
- 6. Sports win probabilities



Live win probabilities

While games are being played, our interactive graphic displays a box for each one that shows updating win probabilities for both teams, as well as the score and the time remaining. These probabilities are derived using **logistic regression** analysis, which lets us plug the current state of a game into a model to produce the probability that either team will win the game.

What day is it today?

BREAKING

Pennsylvania Is This Year's Most Crucial Swing State—Here's How Harris Or Trump Could Still Win Without It

Mary Whitfill Roeloffs Forbes Staff

Mary Roeloffs is a Forbes breaking news reporter covering pop culture.

Follow



0

Oct 25, 2024, 10:14am EDT



NEWSLETTERS SIGN IN

NEWS

CULTURE

MUSIC

PODCASTS & SHOWS

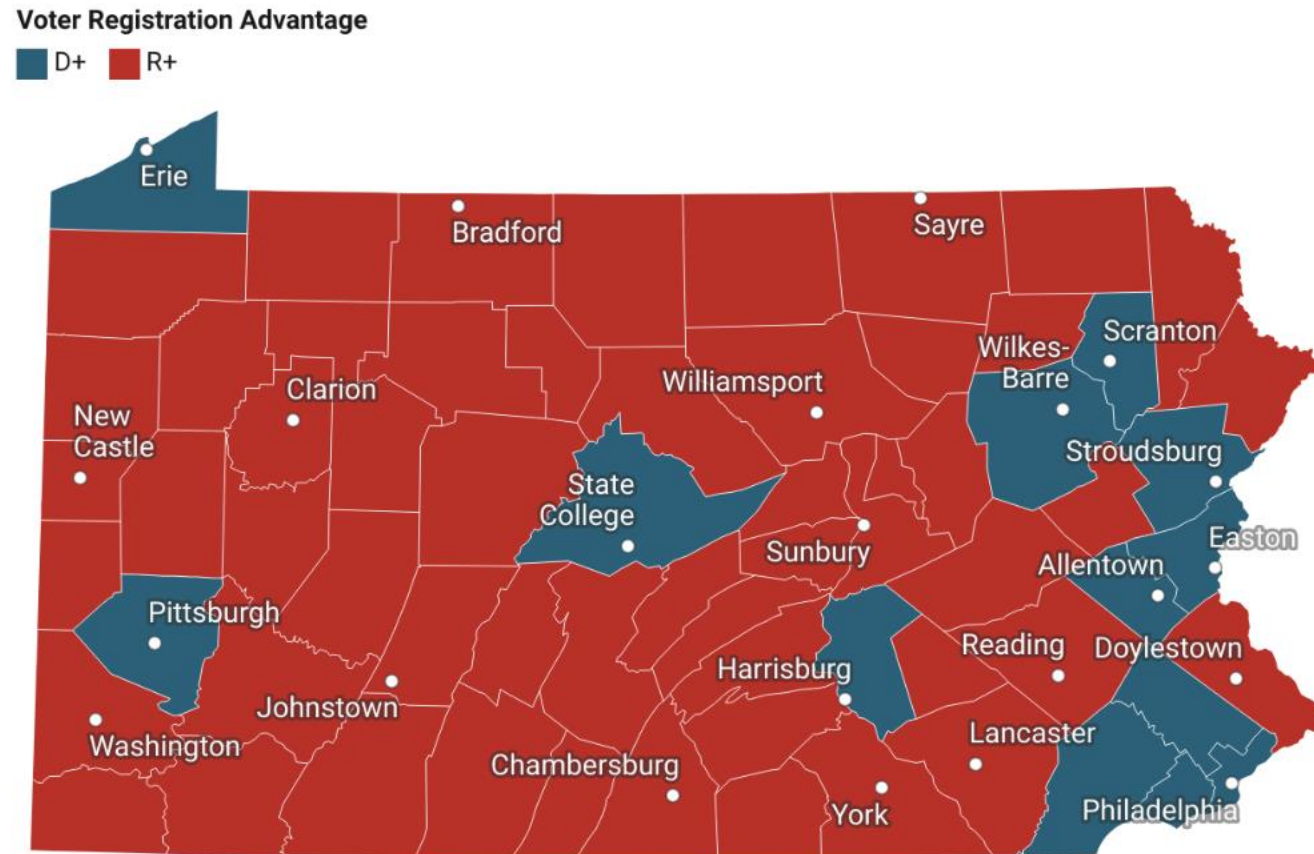
SEARCH

Why Pennsylvania could be the key to the White House

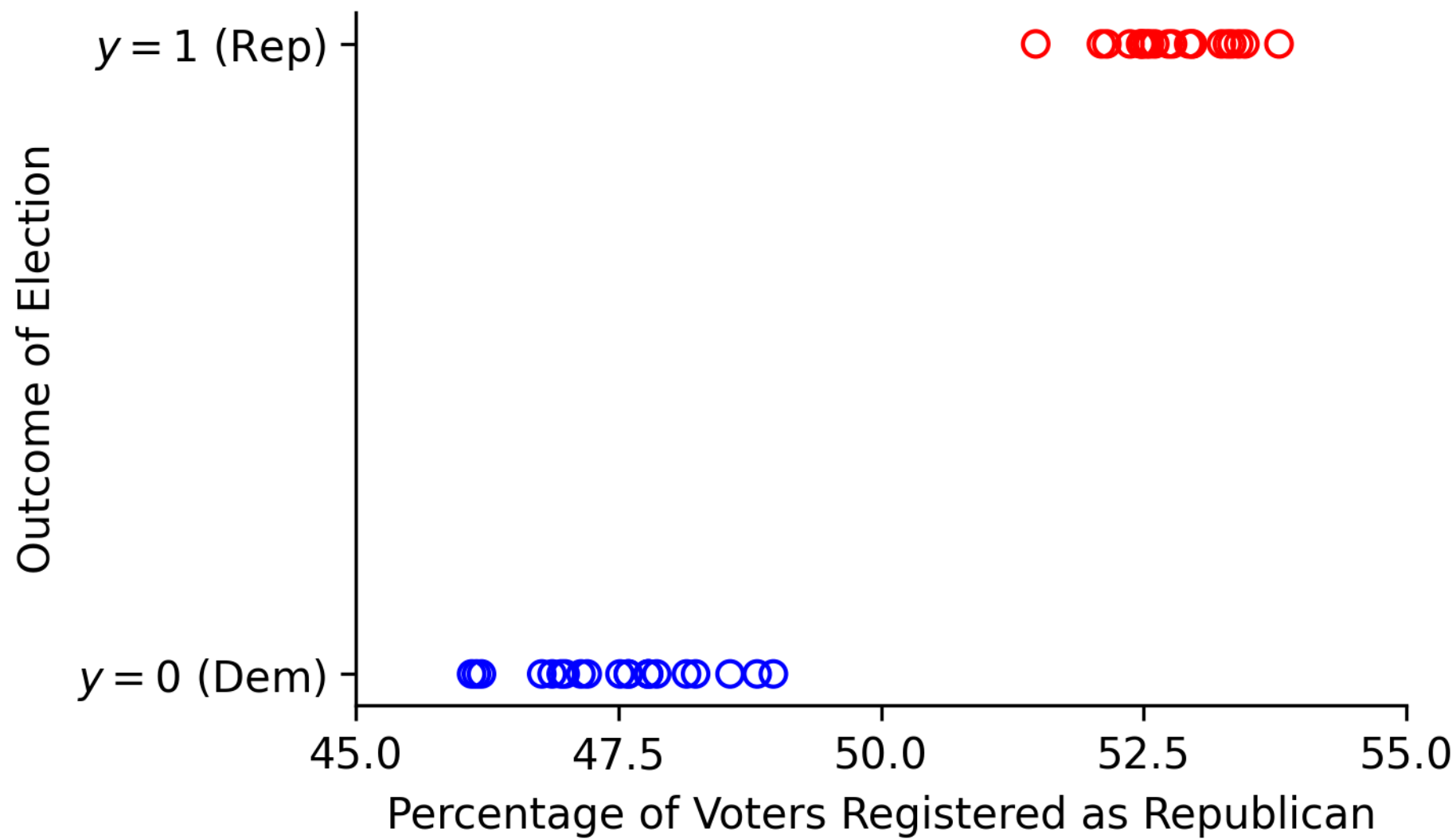
OCTOBER 21, 2024 · 5:00 AM ET

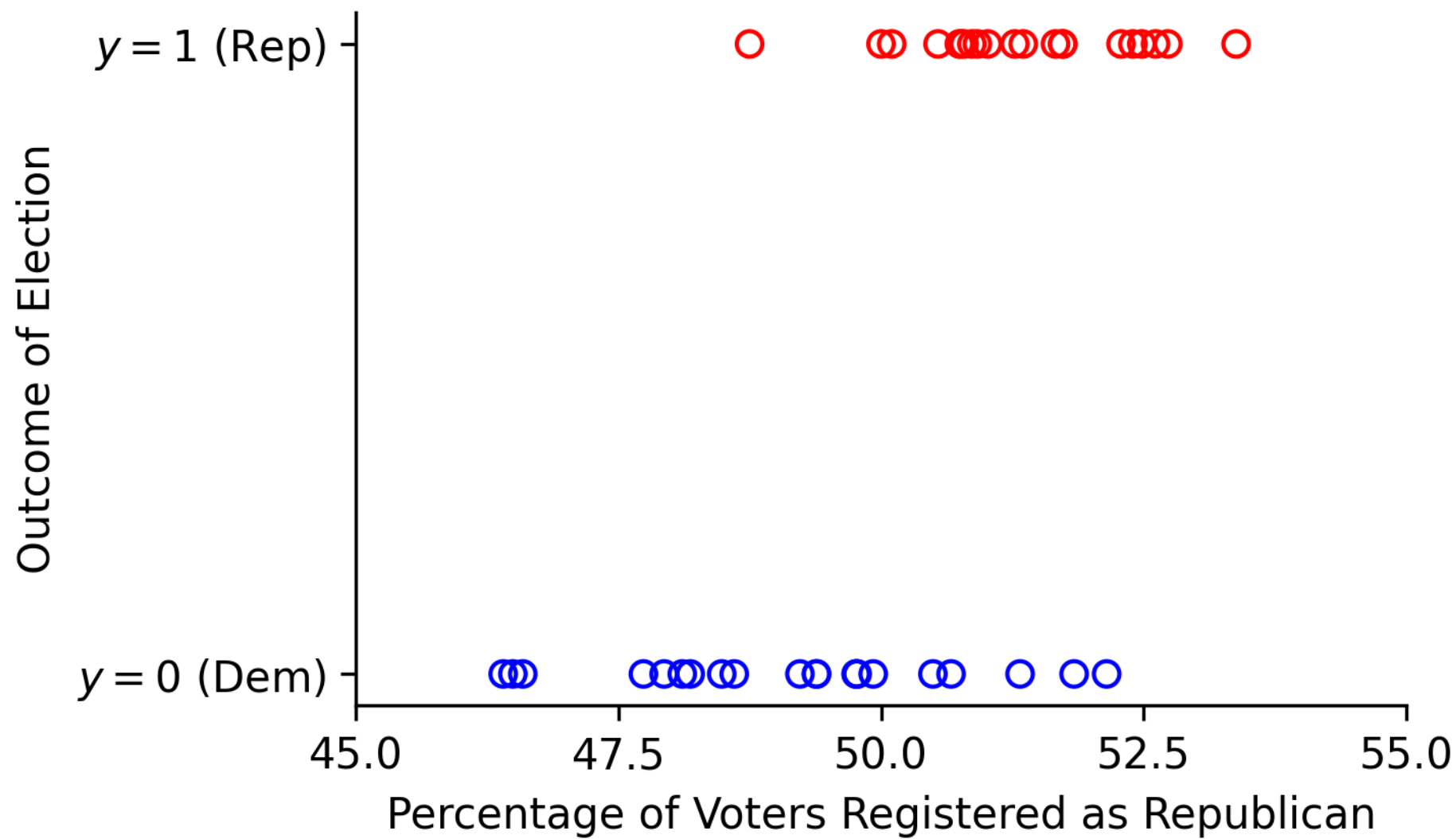
By [Destinee Adams](#), [Obed Manuel](#)

Can we use logistic regression to predict the election result for Union County?



Created with Datawrapper





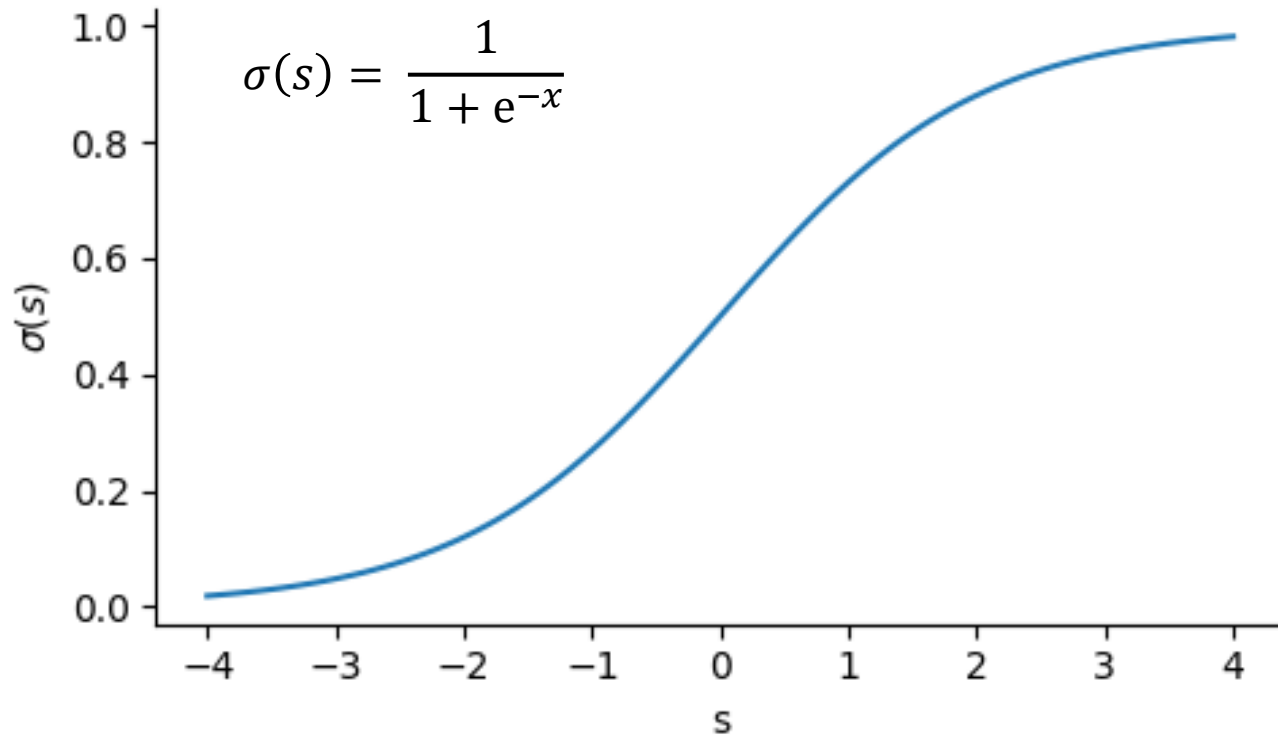
Logistic regression model

- Logistic regression models the *probability* of the outcome ($y = 1$) given the input data x .

$$f(x) = \mathbb{P}(Y = 1 \mid x)$$

$$f(x) = \frac{1}{1 + e^{-(b_0 + b_1 x)}} = \frac{1}{1 + \exp(-(b_0 + b_1 x))}$$

The sigmoid (logistic) function

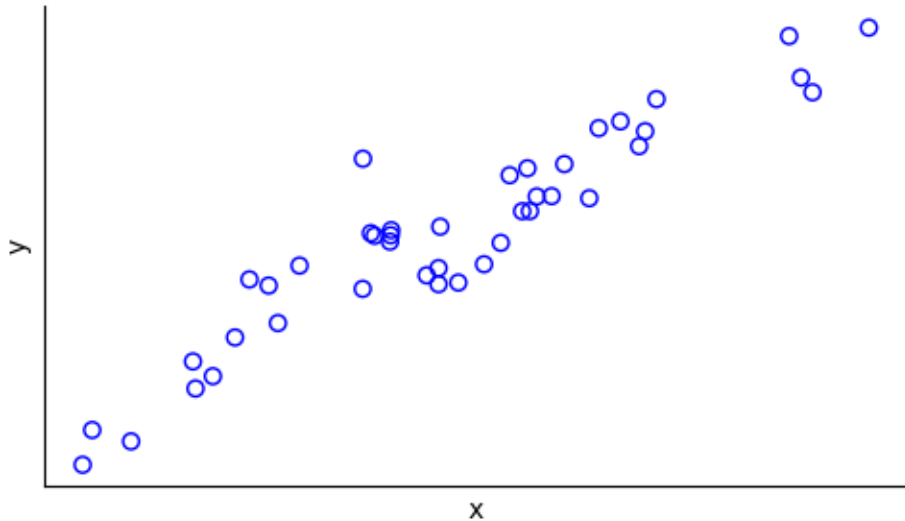


- Transforms a number into a probability between 0 and 1.
- Calculate:
 - $\sigma(0)$
 - $\sigma(100)$
 - $\sigma(-100)$

How do we fit the model?

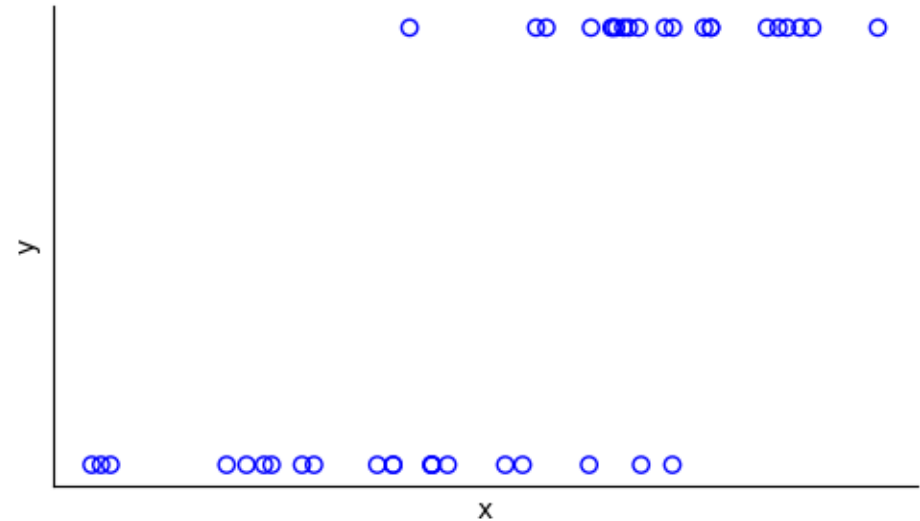
Linear regression

- $y = b_0 + b_1x$
- Find parameters b_0, b_1 for the best-fitting line:



Logistic regression

- $f(x) = \frac{1}{1 + \exp(-(b_0 + b_1x))}$
- Find parameters b_0, b_1 for the best-fitting logistic function:



Follow along in Jupyter Notebook!

In Python:

```
# What does the data look like?
```

```
pa = pd.read_csv("data_wo_union.csv")  
pa
```

	county	percent_rep_voters_registered	vote_2020
0	Adams	57.6	1
1	Allegheny	28.7	0
2	Armstrong	62.0	1
3	Beaver	43.9	1
4	Bedford	73.4	1
...	...	...	...
61	Washington	48.7	1
62	Wayne	58.3	1
63	Westmoreland	51.0	1
64	Wyoming	61.1	1
65	York	51.9	1

66 rows x 3 columns

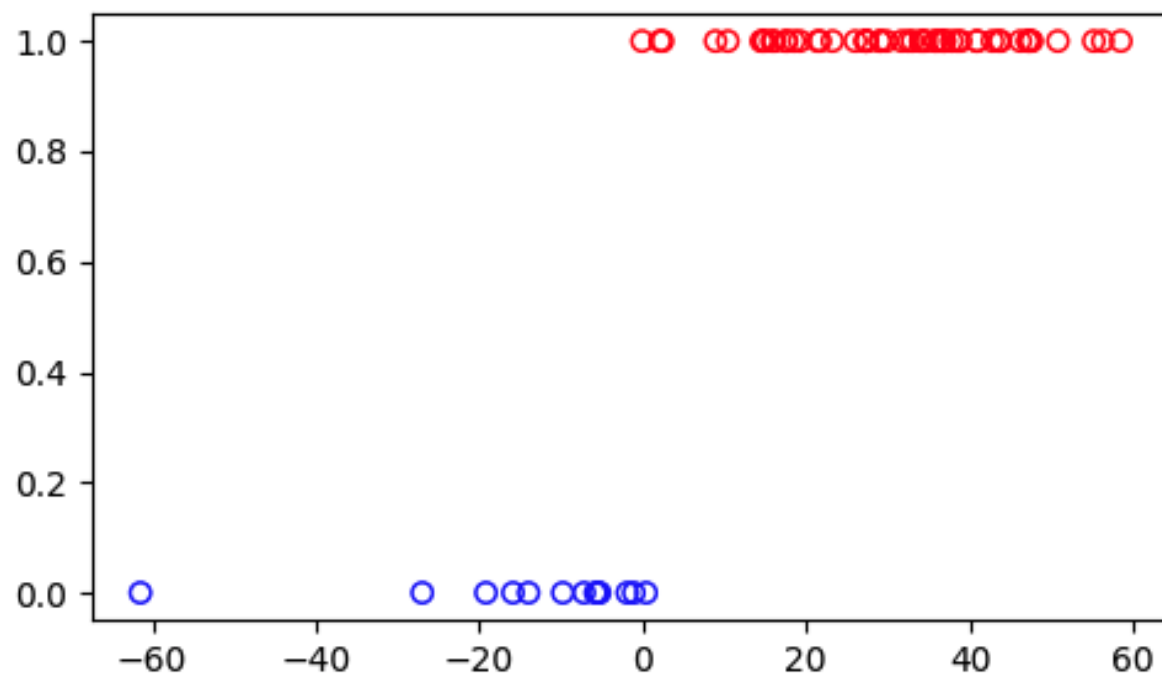
```
# Get X and y
# X = % difference in voting registration
# y = voting result where 1 indicates red (Rep), 0 indicates blue (Dem)
```

```
X = pa.iloc[:,1].to_numpy().reshape(-1,1)
y = pa.iloc[:,2]
```

```
# Plot the data
```

```
fig = plt.figure(figsize=(5,3))
ax = fig.add_subplot(111)
```

```
ax.scatter(X[y==1],y[y==1], marker = 'o', facecolors='none', edgecolors='r')
ax.scatter(X[y==0],y[y==0], marker = 'o', facecolors='none', edgecolors='b')
plt.show()
```



```
# Fit a logistic model, get the coefficient and intercept
```

```
from sklearn.linear_model import LogisticRegression  
clf = LogisticRegression(penalty = 'none').fit(X, y)
```

```
print("Coefficient: ", clf.coef_[0][0])  
print("Intercept: ", clf.intercept_[0])
```

Coefficient: 1.7065090595628383

Intercept: -0.3783588660559433

$$f(x) = \frac{1}{1 + \exp(-(b_0 + b_1 x))}$$

- b_0 is the intercept
- b_1 is the coefficient

```

# Plot the logistic function on top of the data

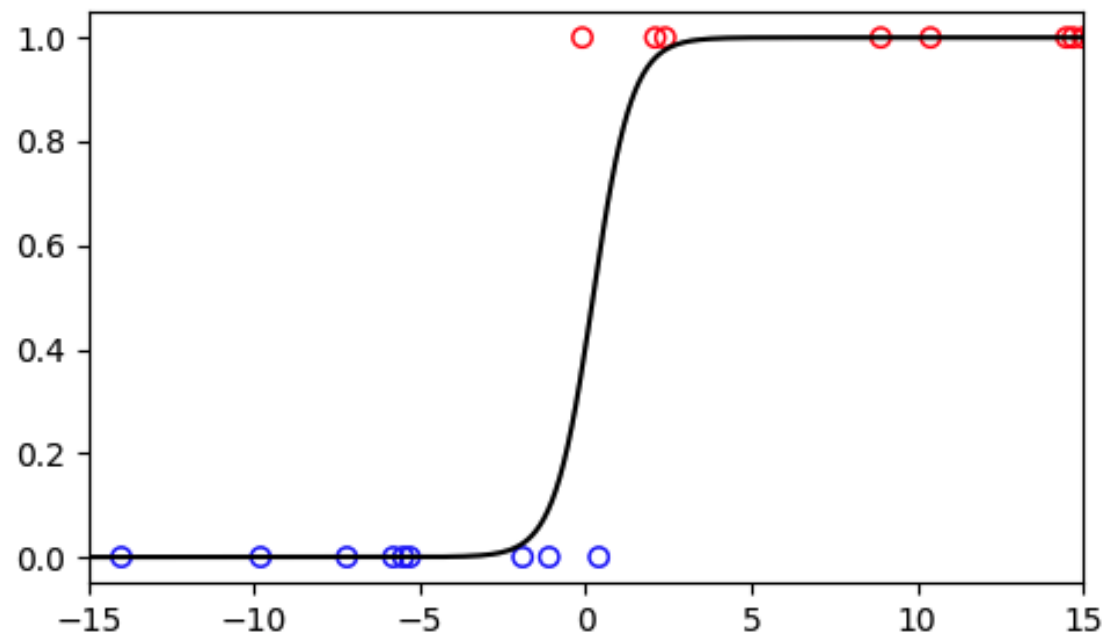
fig = plt.figure(figsize=(5,3))
ax = fig.add_subplot(111)

# Get xs and ys for the logistic function
xs = np.linspace(-60,60,1000)
def f(x):
    return 1/(1+np.exp(-(clf.intercept_[0] + clf.coef_[0][0]*x)))
ys = f(xs)

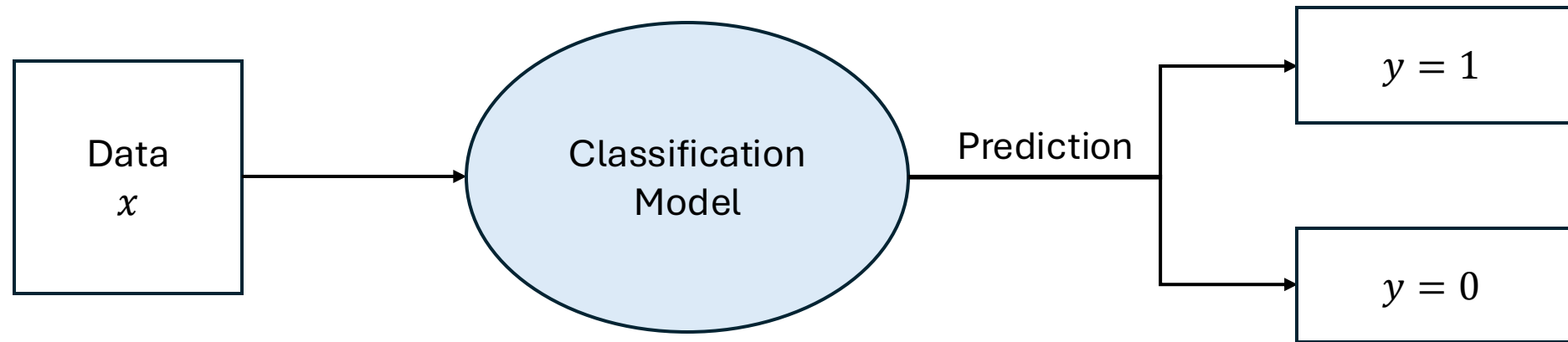
ax.scatter(X[y==1],y[y==1], marker = 'o', facecolors='none', edgecolors='r')
ax.scatter(X[y==0],y[y==0], marker = 'o', facecolors='none', edgecolors='b')
ax.plot(xs, ys, 'black')

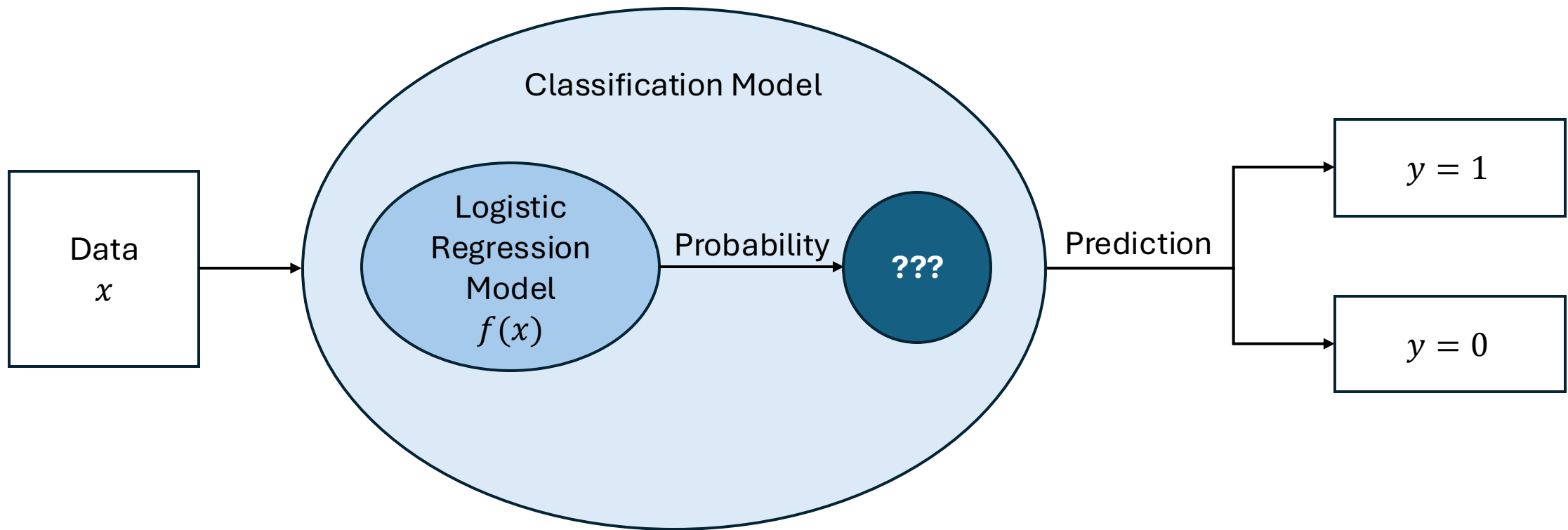
plt.xlim(-15,15)
plt.show()

```



(Binary) Classification





How can we turn logistic regression into a classifier?

New data!

- $x = 0.465$

```
# Get prediction - using Python
```

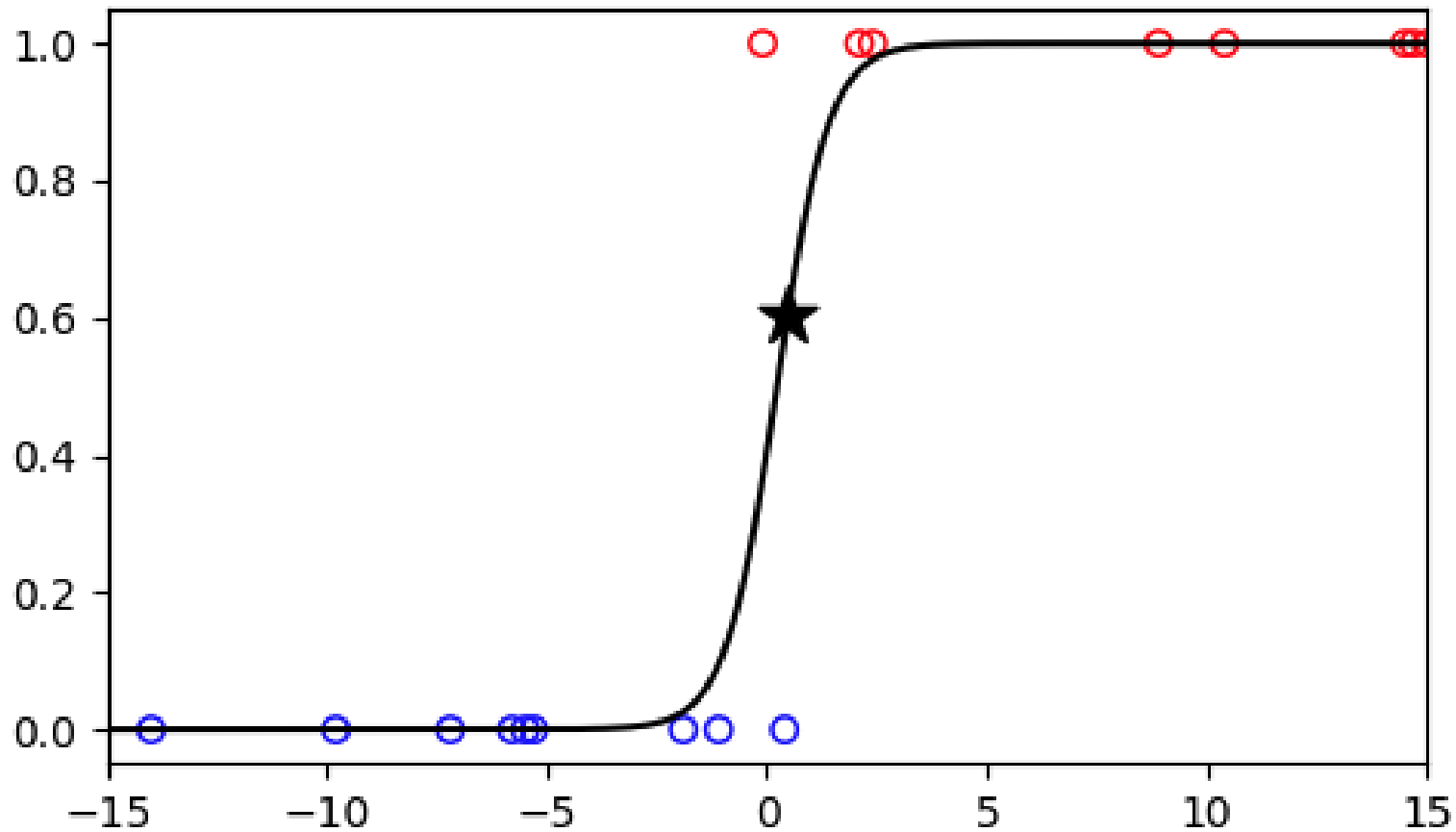
```
clf.predict_proba(np.array(X_test).reshape(-1,1))
```

```
array([[0.39767362, 0.60232638]])
```

```
# Get prediction - using the logistic function directly
```

```
1/(1+np.exp(-(clf.intercept_[0] + clf.coef_[0][0]*X_test)))
```

```
0.6023263817693792
```

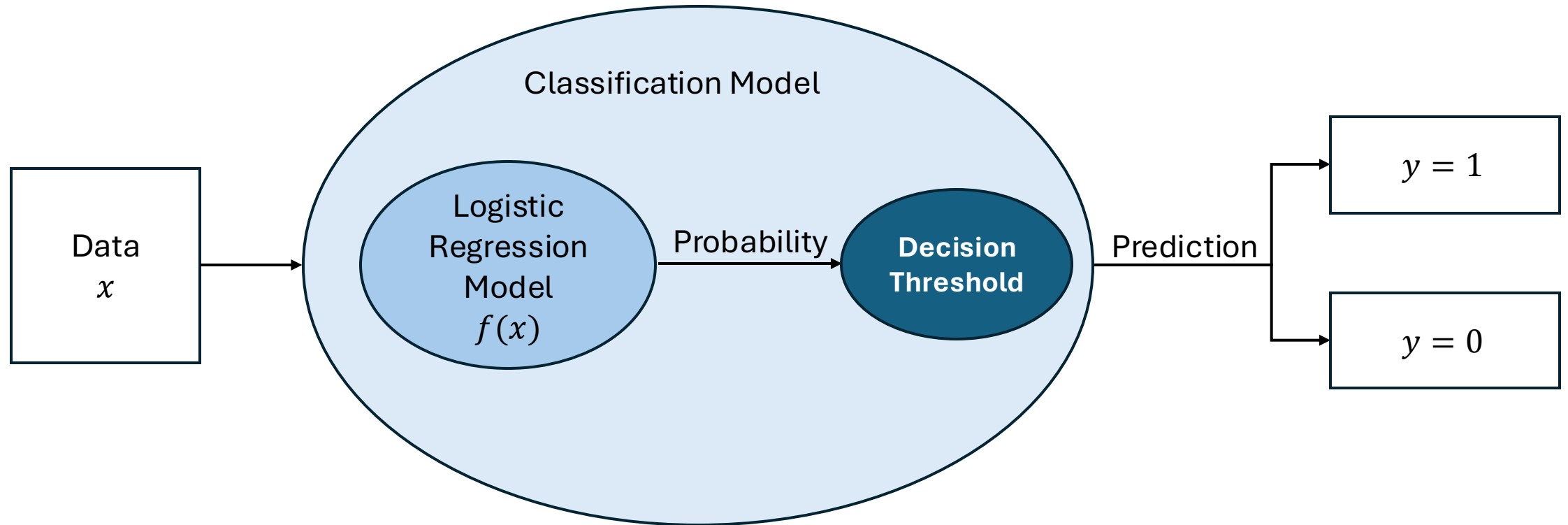


What should we predict as the election result?

What should we predict?

- Predict y based on a **decision threshold** $t = 0.5$:
 - If $f(x) \geq 0.5$, predict $y = 1$
 - If $f(x) < 0.5$, predict $y = 0$
- The *decision boundary* is linear!
- Depending on the problem, other values of t may be better

(Binary) Classification



Preview of more topics in logistic regression

We focused on one predictor variable, but there can be more...

Simple logistic regression

$$f(x) = \frac{1}{1 + \exp(-(b_0 + b_1x))}$$

Multiple logistic regression

$$f(x) = \frac{1}{1 + \exp(-(b_0 + b_1x_1 + b_2x_2 + \cdots + b_{100}x_{100}))}$$

Preview of more topics in logistic regression

We focused on binary classification, but we can do multi-class classification...

Simple logistic regression

$$f(x) = \frac{1}{1 + \exp(-(b_0 + b_1 x))}$$

Multinomial logistic regression

$$f_k(x) = \frac{\exp(b_0^k + b_1^k x)}{\sum_{j=1}^K \exp(b_0^j + b_1^j x)}$$

A quick
aside...

2024 Nobel Prize

2024 Nobel Prize in physics
awarded to John J. Hopfield,
Geoffrey E. Hinton for
discoveries that 'enable
machine learning with artificial
neural networks'



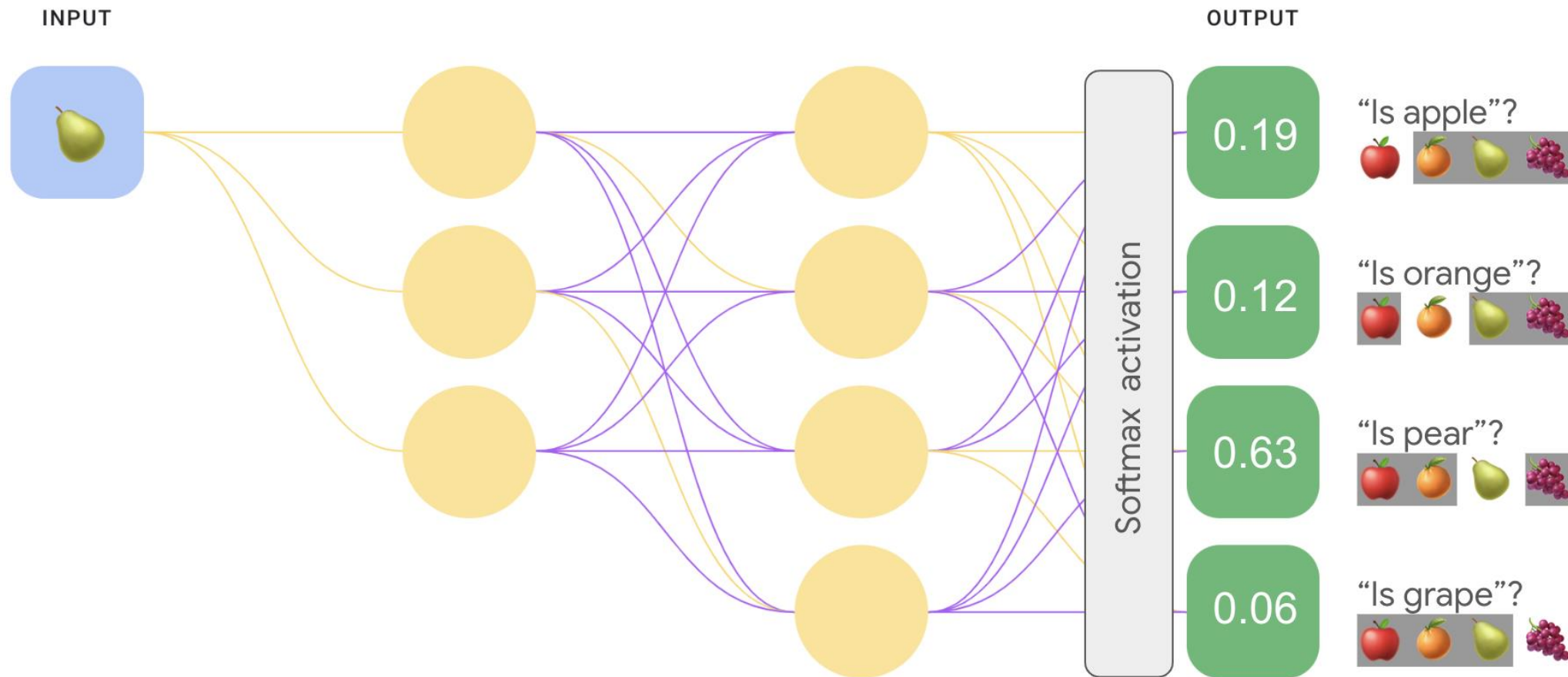
John J.
Hopfield



Geoffrey E.
Hinton

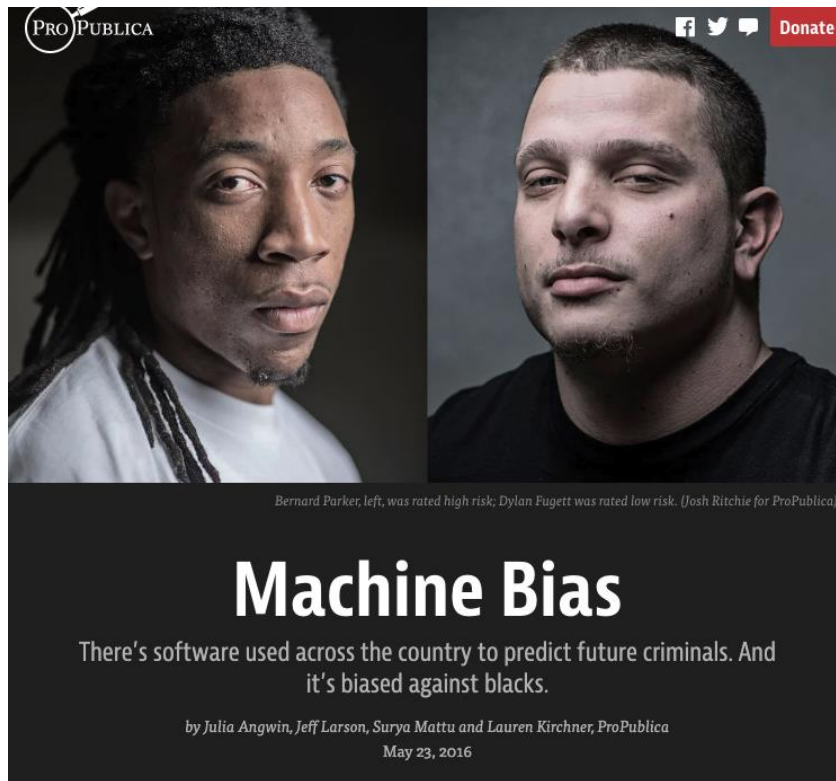


- Neural networks often use a version of the sigmoid function!
- Transforms numbers into class **probabilities**



Preview of more topics in logistic regression

Logistic regression is an *interpretable* model: we can interpret the coefficients.



PRO PUBLICA

Facebook Twitter Messenger Donate

Bernard Parker, left, was rated high risk; Dylan Pugett was rated low risk. (Josh Ritchie for ProPublica)

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica
May 23, 2016

How We Analyzed the COMPAS Recidivism Algorithm

by Jeff Larson, Surya Mattu, Lauren Kirchner and Julia Angwin
May 23, 2016

Risk of General Recidivism **Logistic Model**
Dependent variable:
Score (Low vs Medium and High)

Female	0.221*** (0.080)
Age: Greater than 45	-1.356*** (0.099)
Age: Less than 25	1.308*** (0.076)
Black	0.477*** (0.069)
Asian	-0.254 (0.478)
Hispanic	-0.428*** (0.128)
Native American	1.394* (0.766)
Other	-0.826*** (0.162)
Number of Priors	0.269*** (0.011)
Misdemeanor	-0.311*** (0.067)
Two year Recidivism	0.686*** (0.064)

We used those factors to model the odds of getting a higher COMPAS score. According to [Northpointe's practitioners guide](#), COMPAS "scores in the medium and high range garner more interest from supervision agencies than low scores, as a low score would suggest there is little risk of general recidivism," so we considered scores any higher than "low" to indicate a risk of recidivism.

Our **logistic** model found that the most predictive factor of a higher risk score was age. Defendants younger than 25 years old were 2.5 times as likely to get a higher score

Today, we learned...

- Examples of logistic regression in the world
- What is logistic regression and what does the model look like?
- How can we perform classification using logistic regression?

What will you predict?

